1. Tasa mensual real del outcome

Yo empezaría por esta, porque es la más granular y después podemos consolidarla por año.

```
monthly_population_outcome_rate <- sdf_sql(
  sc,
  "
  WITH population AS (

      /* Full population at each monthly observation date */
      SELECT DISTINCT
          party_id,
          data_date AS observation_date

      FROM bc3.person_restricted

      WHERE data_date >= '2022-01-01'
        AND data_date <= '2025-12-31'
        AND data_date = LAST_DAY(data_date)
  ),

  at_risk_population AS (

      SELECT
          p.party_id,
          p.observation_date,
          f.first_qualifying_alert_date

      FROM population p

      LEFT JOIN bu_frd.party_first_valid_alert_all f
          ON p.party_id = f.party_id

      /* Party must not already have experienced the first qualifying alert */
      WHERE
          f.first_qualifying_alert_date IS NULL
          OR f.first_qualifying_alert_date > p.observation_date
  ),

  labeled AS (

      SELECT
          party_id,
          observation_date,

          CASE
```

```
            WHEN first_qualifying_alert_date > observation_date
             AND first_qualifying_alert_date
                 <= ADD_MONTHS(observation_date, 3)
            THEN 1
            ELSE 0
          END AS label_3m

      FROM at_risk_population
  )

  SELECT
      observation_date,

      YEAR(observation_date)  AS observation_year,
      MONTH(observation_date) AS observation_month,

      COUNT(*) AS eligible_observations,

      SUM(label_3m) AS label_1,

      COUNT(*) - SUM(label_3m) AS label_0,

      SUM(label_3m) * 1.0
          / COUNT(*) AS outcome_rate,

      100.0 * SUM(label_3m)
          / COUNT(*) AS outcome_rate_pct

  FROM labeled

  GROUP BY
      observation_date,
      YEAR(observation_date),
      MONTH(observation_date)

  ORDER BY observation_date
  "
) %>%
  collect()

monthly_population_outcome_rate
```

El output debería tener 48 filas:

observation_date	year	month	eligible	label_1	label_0	outcome_rate_pct

2022-01-31	2022	1	...	...	...	...
2022-02-28	2022	2	...	...	...	...
...	...	...	...	...	...	...
2024-05-31	2024	5	...	...	...	...
...	...	...	...	...	...	...
2025-12-31	2025	12	...	...	...	...

---

2. Tasa anual real del outcome

Para la tasa anual utilizaría **las mismas observaciones mensuales**, pero agregadas por año:

```
annual_population_outcome_rate <- sdf_sql(
  sc,
  "
  WITH population AS (

      SELECT DISTINCT
          party_id,
          data_date AS observation_date

      FROM bc3.person_restricted

      WHERE data_date >= '2022-01-01'
        AND data_date <= '2025-12-31'
        AND data_date = LAST_DAY(data_date)
  ),

  at_risk_population AS (

      SELECT
          p.party_id,
          p.observation_date,
          f.first_qualifying_alert_date

      FROM population p

      LEFT JOIN bu_frd.party_first_valid_alert_all f
          ON p.party_id = f.party_id

      WHERE
          f.first_qualifying_alert_date IS NULL
```

```
        OR f.first_qualifying_alert_date > p.observation_date
  ),

  labeled AS (

    SELECT
      party_id,
      observation_date,

      CASE
        WHEN first_qualifying_alert_date > observation_date
         AND first_qualifying_alert_date
             <= ADD_MONTHS(observation_date, 3)
        THEN 1
        ELSE 0
      END AS label_3m

    FROM at_risk_population
  )

  SELECT
    YEAR(observation_date) AS observation_year,

    COUNT(*) AS eligible_observations,

    SUM(label_3m) AS label_1,

    COUNT(*) - SUM(label_3m) AS label_0,

    SUM(label_3m) * 1.0
      / COUNT(*) AS outcome_rate,

    100.0 * SUM(label_3m)
      / COUNT(*) AS outcome_rate_pct

  FROM labeled

  GROUP BY YEAR(observation_date)

  ORDER BY observation_year
  "
) %>%
  collect()

annual_population_outcome_rate
```

El resultado será:

Year	Eligible Observations	Label 1	Label 0	Outcome Rate %
2022	...	...	...	...%
2023	...	...	...	...%
2024	...	...	...	...%
2025	...	...	...	...%

Una precisión importante sobre esos números

Esta tasa anual será una **annual observation rate**, porque un mismo party puede contribuir en varios `observation_date` mientras siga estando at risk.

Eso es consistente con la pregunta que estamos haciendo:

> De todas las oportunidades mensuales en las que una persona podía ser observada, ¿qué proporción presentó su first qualifying alert dentro de los siguientes tres meses?

Por ejemplo, si una persona tiene su first qualifying alert en junio de 2024, puede ser:

- `label_3m=1` en marzo;
- `label_3m=1` en abril;
- `label_3m=1` en mayo.

Eso es correcto para nuestro **3-month prospective outcome**, aunque exista solamente un first alert real.

Por eso **no esperes que `label_1` anual de este query sea igual a 2,434,797 en 2024**. Ese número es el conteo de personas cuyo `first_qualifying_alert_date` ocurrió en 2024; aquí estamos contando **observation opportunities cuyo futuro de 3 meses contiene ese evento**.

Esta es precisamente la tasa que mejor se alinea con la definición temporal de `label_3m`.

Yo correría primero `monthly_population_outcome_rate` y después `annual_population_outcome_rate`. Cuando me pases esos dos resultados, ya podemos interpretar directamente las tasas de 2022–2025 y especialmente qué ocurre antes y durante el pico de 2024.